Spotify Track Popularity Analysis

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Our Data

Source

We found our specific dataset on Kaggle, which was created using the Spotify API. The dataset we are using includes data on approximately 8,500 songs released in 2025.

Structure

This dataset includes variables such as track_popularity, artist_name, artist_popularity, artist_followers, as well as whether or not the track is explicit.

Our data science pipeline

1

Research Questions

Specify some research questions in order to guide our EDA and modeling, select one that best represents our interests in analyzing the data

2

EDA

Generate summary statistics and basic charts for variables of interest, to better understand our data

3

Run and select supervised model

Run several types of supervised models, compare results and select which one best serves our needs

4

Run and select unsupervised model

Run several types of unsupervised models, compare results and select which one best serves our needs

5

Make conclusions based on results of models

Interpret our best models in the context of the research question, generate hypothetical predictions, give our conclusive (or inconclusive) response to the research question

01

Research Questions

Research Questions

In order to move forward in our analysis, we need a research question to guide our analysis. We want our question to:

- Specify a response variable to which we can make inferences
- Make clear which features are of interest to us
- Be (hypothetically) answerable in the context of machine-learning methods
- Ask an question that would be useful and/or interesting to have an answer to

Potential questions:

Are popular artists more likely to release explicit songs, and is this correlated to the track's popularity?

Classifying songs as 'high' or 'low' popularity, do we see clustering patterns that reflect differences in track-level and artist-level characteristics?

How do track-level and artist-level characteristics relate to a song's popularity, and which features show the strongest relationships with track popularity?

02

Exploratory Data Analysis

Exploratory Data Analysis

Summary statistics for the track_popularity variable (our response variable):

- Mean: 52.36
- Standard Deviation: 23.82
- Range: 0-99
- 25th percentile: 39
- Median: 58
- 75th percentile: 71

Correlation analysis results

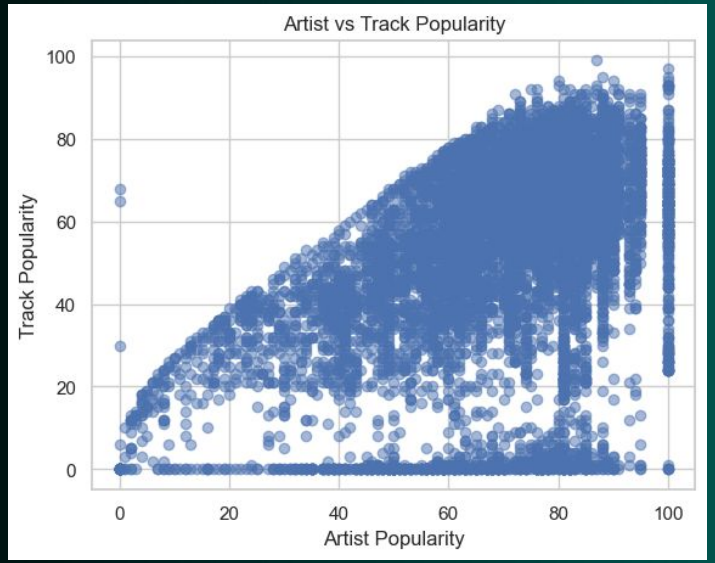
Artist_popularity: moderate positive correlation

Artist_followers: weak positive correlation

Track_duration_min: weak positive correlation

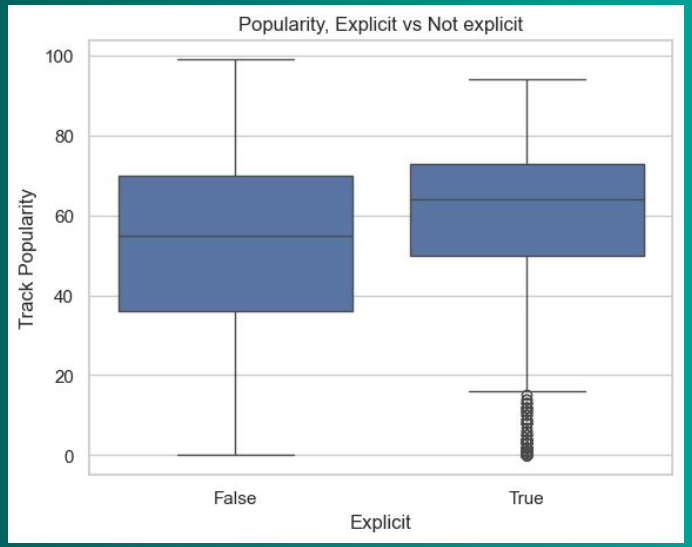
Album_total_tracks: weak positive correlation

Exploratory Data Analysis: Visuals



- Heteroskedastic trend
- Visually looks correlated

25% of tracks are explicit, while the remaining 75% are not.



- Slight difference in median and percentiles

03

Supervised Modeling

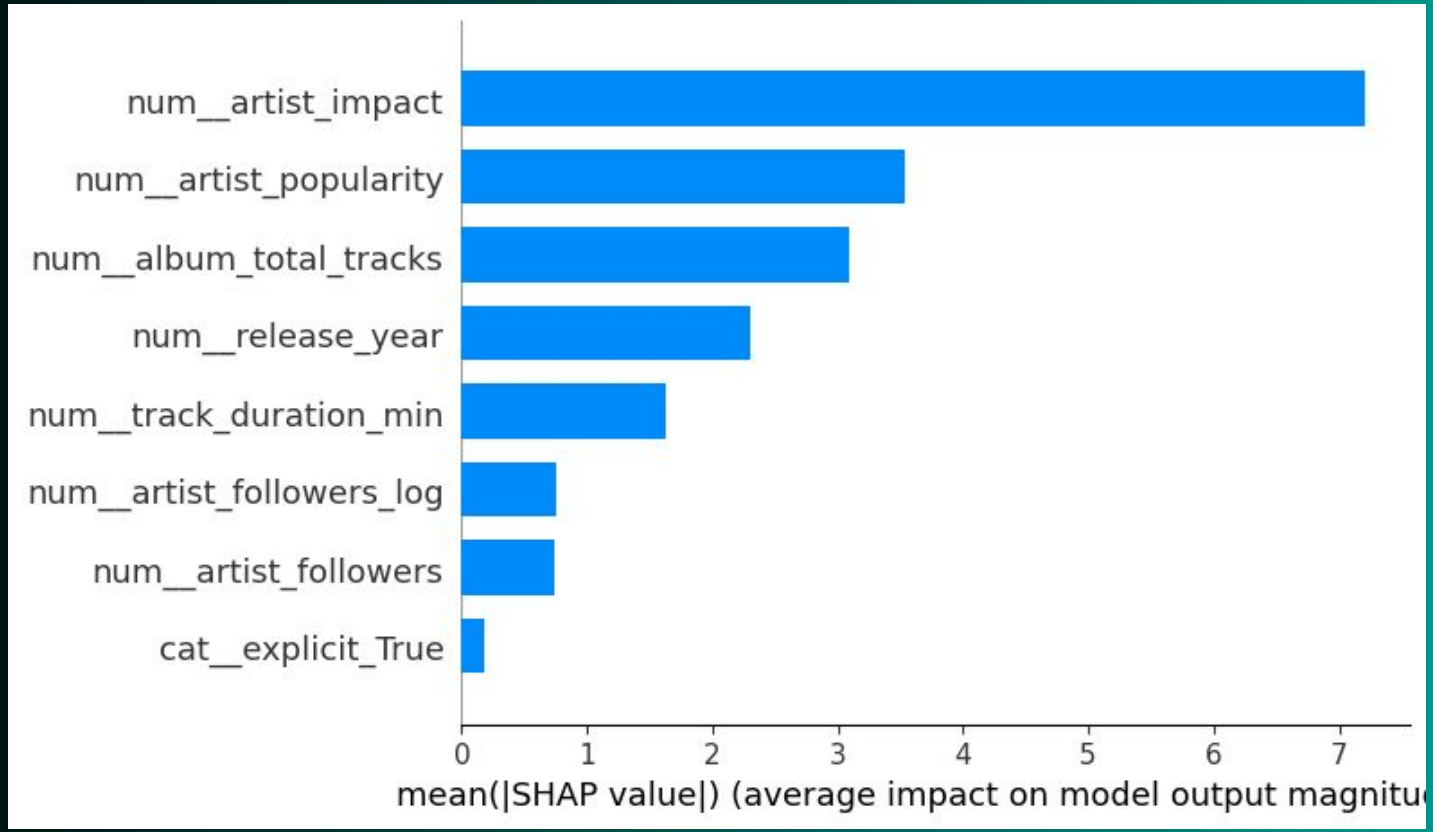
Supervised Modeling Approach

- Feature Engineering
- Preprocessing pipeline (imputation, scaling, and encoding)
- 80/20 train-test split
- 5-fold cross-validation (GridSearchCV)
- Evaluation metrics: RMSE and R-squared

Models Compared

Model	Key Hyperparameters	Test RMSE	Test R^2
Linear Regression	None	20.73	0.277
Ridge Regression	alpha $\in$ {0.01, 0.1, 1, 10, 100}	20.73	0.277
Random Forest	n_estimators, max_depth, min_samples_split, min_samples_leaf	**18.85**	**0.402**
Gradient Boosting	learning_rate, max_depth, max_iter	19.19	0.380
SVR	C, gamma	20.79	0.273

Feature Importance (SHAP)



Key Findings

- Artist-level characteristics are the stronger predictors of popularity
- Track-level features provide smaller, secondary contributions
- The Model explains moderate variation (R-squared around .40)

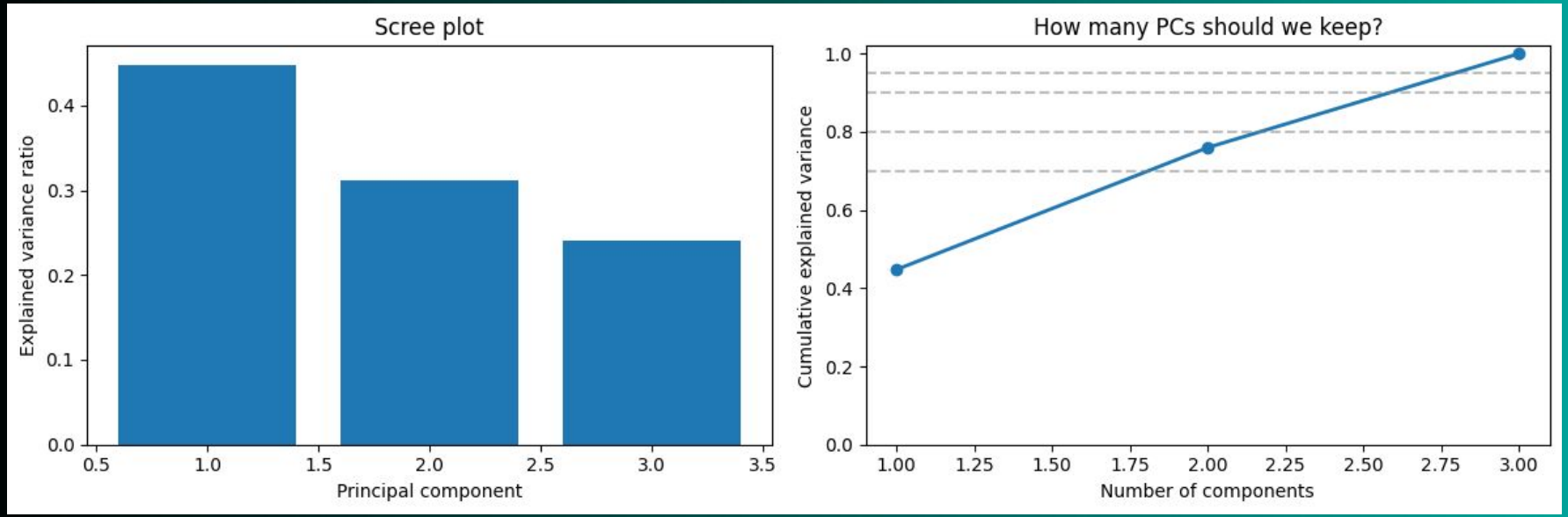
- Highly skewed target (many low-popularity tracks)
- Noise limits predictive performance
- Missing features

04

Unsupervised Modeling

Principal Component Analysis

PCA verifies our 3 variables chosen for clustering (Artist impact, track duration, album tracks)

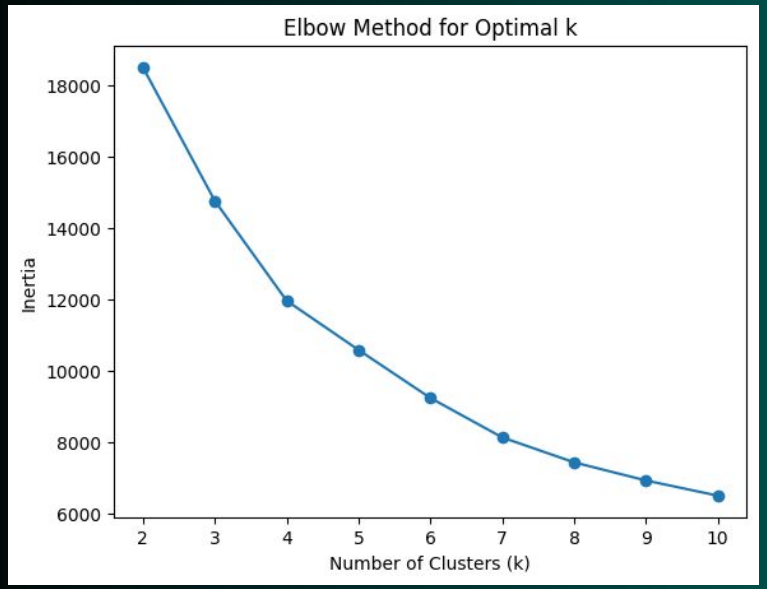


These three significantly contribute to the top two principal components

Accounts for 76% of variation in the response

k-Means Clustering

Optimal clusters k not obvious from elbow method

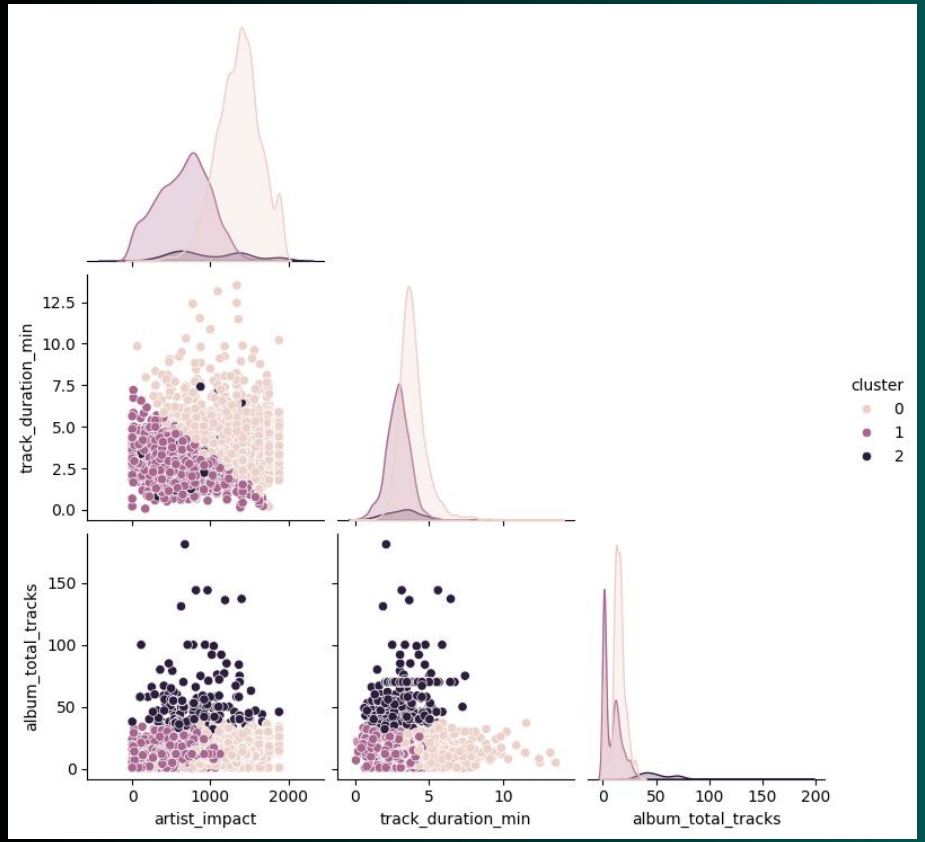


3-4 clusters looks reasonable, but not definitive

Highest silhouette score is at k=3

We will be using k=3 for our clustering analysis

k-Means Clustering, cont.



Characteristics of the 3 clusters

- Cluster 0: High impact artists, high popularity tracks
- Cluster 1: Low impact artists, lower popularity tracks
- Cluster 2: Very large album size, low popularity

Disparate # of observations in each cluster

- Cluster 0 : 5129 tracks
- Cluster 1: 3077 tracks
- Cluster 2: 376 tracks

Supports findings from supervised model

- Artist level characteristics drive popularity more than track level characteristics
- Some track level characteristics contribute to popularity, but are less important

Conclusions

Artist level characteristics are most important in explaining song popularity

- Popular artists with more followers tend to have much more popular songs

Track level characteristics are less important in explaining song popularity, but are more within the control of artists

- Longer songs tend to be a bit more popular
- Songs in larger albums tend to be a bit less popular

Much of the variance in track popularity is still unexplained

- Future analyses could benefit from datasets with additional features (genre, song tempo, instruments used)

Thank You

Questions?